

512Mbits (x8) 528 Bytes Page,
NAND Flash Memory
Verilog HDL Model User Manual

The provided files description:

NAND512R3A.v

NAND512R3A Verilog behavior model.

NAND_PARAM.v

Defines parameters and DC/AC characteristics of NAND512R3A.

NAND_TEST.v

Stimuli Example for NAND512R3A device.

NAND_BLK_INIT.txt

An example of the memory block initialization file.

HOW TO USE THE INITIALIZATION FILE

An initialization file (NAND_BLK_INIT.txt) is offered to define the initialization content of the memory array before starting a simulation. This file initializes a block of the NAND Flash every time.

When starting a simulation, the whole simulated memory array is automatically loaded with the contents of the NAND_BLK_INIT.txt file one block by one block.

The NAND_BLK_INIT.txt file provided with this Verilog HDL model is filled with "FF" but it can be easily replaced by another user's initialization file.

User Control Interface:

The user can control the number of blocks and AC timing characteristics by setting the parameters in the models.

Parameters of AC timing characteristics are defined in NAND_PARAM.v file.

The number of blocks is controlled by macro "MEMORY_ALLOC" and "NO_OF_BLK_ALLOC" in NAND512R3A.v file.

NO_OF_BLK_ALLOC:

1: all blocks of NAND Flash is selected.

0: partial blocks of NAND Flash is selected.

NO_OF_BLK_ALLOC:

This macro is used to select the number of available blocks in NAND Flash, when "NO_OF_BLK_ALLOC" is "0".

DRIVER FILES DESCRIPTION

NAND_TEST.v provide an example for driving the NAND512R3A Flash memory model. It is easy to replace this example file by user's NAND Flash memory stimuli file.

In this example, following tasks are defined. These tasks are easy to be invoked to generate the stimuli for memory access.

READ_A: Generates a Read Area A Instruction.

READ_B: Generates a Read Area A Instruction.

READ_C: Generates a Read Area C Instruction.

RESET: Generates a Reset Instruction.

READ_ES: Generates a Read Electronic Signature Instruction sequence.

READ_SR: Generates a Read Status Register Instruction sequence.

READ: Generates a Read Memory Instruction Sequence.

Format: *READ (cmd, addr_1,addr_2,addr_3,addr_4,n)*

cmd: Read Command Code; 00H, 01H or 50H

addr_1: 1st cycle of input address (A0~ A7) for read

addr_2: 2nd cycle of input address(A9~ A16) for read

addr_3: 3rd cycle of input address(A17~ A24) for read

addr_4: 4th cycle of input address(A25) for read

n: the number of data bytes to be read out.

PROGRAM: Generates a Page Program Instruction Sequence.

Format: *PROGRAM(addr_1,addr_2,addr_3,addr_4,data,n)*

addr_1: 1st cycle of input address (A0~ A7) for program

addr_2: 2nd cycle of input address(A9~ A16) for program

addr_3: 3rd cycle of input address(A17~ A24) for program

addr_4: 4th cycle of input address(A25) for program

data: the data byte to be written at each address.

n: the number of data bytes to be written.

COPY_BACK: Generates a Copy Back Program Instruction Sequence.

Format: *COPY_BACK(address_src, address_tgt)*

address_src: source page address (A0~ A7,A9~ A25) for copy back program

address_tgt: destination page address (A0~ A7,A9~ A25) for copy back program

BLOCK_ERASE: Generates a Block Erase Instruction Sequence.

Format: *BLOCK_ERASE(addr_1,addr_2,addr_3)*

addr_1: 1st cycle of input block address (A9~ A16) for erase

addr_2: 2nd cycle of input block address(A17~ A24) for erase

addr_3: 3rd cycle of input block address(A25) for erase

REVISION HISTORY

Table 1. Document Revision History

Date	Version Revision	Details
17-Dec-2004	1.0	First Issue